

TrES-3b

R-2807

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130801 · created 2026-09-05T00:56:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456505.6978 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.057
Transit depth (Rp/Rs)^2	3.62 +/- 2.18 [%]
Orbital Inclination (inc)	81.07 +/- 2.83
Airmass coefficient 1 (a1)	0.058 +/- 0.022
Airmass coefficient 2 (a2)	2.68 +/- 1.29
Scatter in the residuals of the lightcurve fit is	41.15 %
Best Comparison Star	#2 - [146, 301]
Optimal Aperture	6.42
Optimal Annulus	11.13
Transit Duration (day)	0.043 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.057 vs 0.16309 (deviation 0.38σ)
Duration fit vs published	0.043 d vs 0.0567 d (deviation 0.38σ)
Mid-time O–C	-1.0 min
Depth SNR	2.7
Residual scatter	41.15 %

Light curve

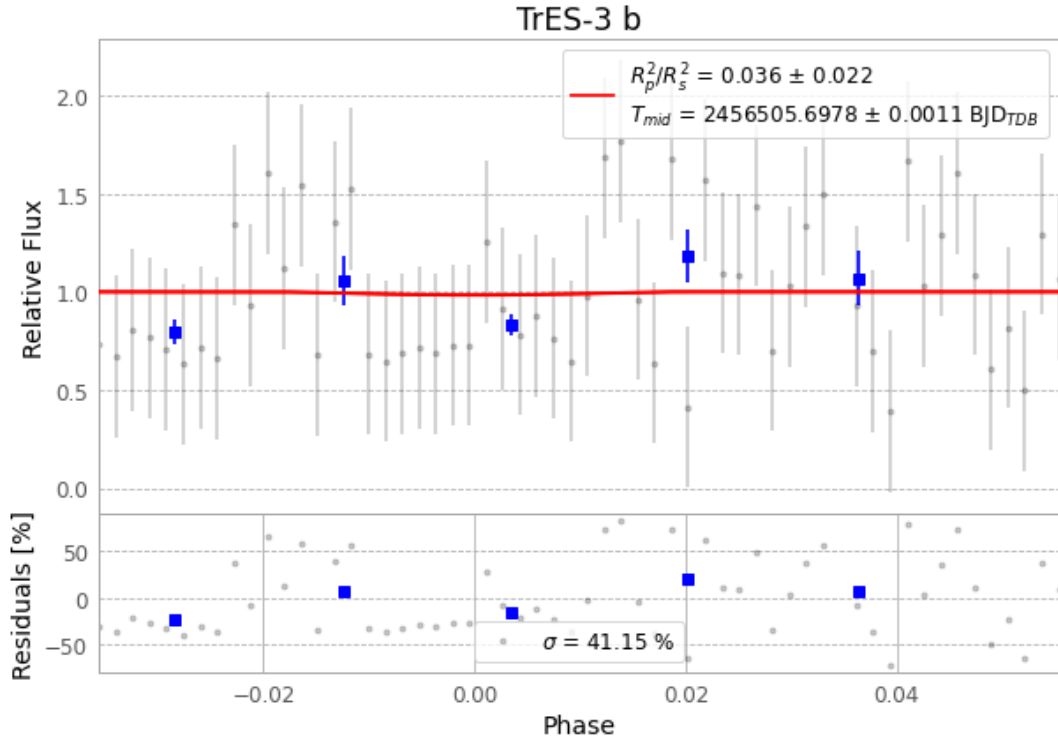


Figure 1 — FinalLightCurve_TrES-3 b_2013-07-31.png (SHA-256 f396940f093d34d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [481, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fa28da5b976cb80a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), TRES-3130801033312.FITS → TRES-3130801062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130801.log (2c1adfa3bf26...), FinalLightCurve_TrES-3 b_2013-07-31.png (f396940f093d...), FinalLightCurve_TrES-3 b_2013-07-31.csv (fb57de30f2b2...)

Compute resources

Wall time 370.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 00:56Z.